This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2018.2841806, IEEE Transactions on Automatic Control

On the Local Stabilization of Hybrid Limit Cycles in Switched Affine Systems

Mohammed Benmiloud, Atallah Benalia, Mohamed Djemai, *Senior Member, IEEE,* and Michael Defoort

***Abstract*—Most of the proposed controllers for switched affine systems are able to drive the system trajectory to a sufficiently small neighborhood of a desired state. However, the trajectory behavior in this neighborhood is not generally considered despite the fact that the system performance may be judged by its steady operation as the case of power converters. This note investigates the local stabilization of a desired limit cycle in switched affine systems using hybrid Poincaré map approach. To this end, interesting algebraic properties of the Jacobian of the hybrid Poincaré map are firstly discussed and used for the controller design to achieve asymptotic stability conditions of the limit cycle. A DC-DC four-level power converter is considered as an illustrative example to highlight the developed results.**

***Index Terms*—Switched affine system, limit cycle, stabilization, hybrid Poincaré map, multilevel power converter**

I. Introduction

During the past decades, switched dynamical systems have witnessed remarkable developments from theoretical point of view to practical insights with potential benefits to many real world applications in nature, engineering, and social sciences [1]. A switched system consists of a family of continuous-time subsystems and a rule that orchestrates the switching among them [2]. The stabilizing switching law design for such class of systems represents a challenging issue notably in the case where the trajectory of the system has to be driven to a desired point, which does not belong to the set of subsystem equilibria [3], [4]. In this case, the switching law compels the system trajectory to converge to a neighborhood of the desired state and remains there.

Several research studies have investigated the stabilization problem of switched systems with different approaches [1]–[7]. For example, the authors in [7] have tackled this issue for switched affine systems by looking for the existence of a Hurwitz convex combination where the stability of the desired state, named switched equilibrium point, can be guaranteed using a quadratic Lyapunov function. Another approach based on state space partition has been developed in [4] by solving a bilinear matrix inequality to guarantee the practical stability of a desired state. In addition, the local stabilization of the latter has been addressed in [3] where the main idea is based on an alternative representation of the switched system as a nonlinear system with input constraints. The proposed switching law is derived by emulating locally classical controllers. Furthermore, optimal and predictive control schemes for switched systems have been discussed in [8], [9]. Although these strategies achieve the control requirements with different performances, the steady state behaviour of the system trajectory in the neighborhood of the desired state is not generally directly controlled despite the fact that it may affect considerably the system performance.

Power converters are good examples of switched systems where the desired state is not an equilibrium of any subsystem and due to technological limitations some physical constraints have to be considered such as: maximum switching rate, dwell time and succession of subsystems. As a remedy, some research studies have discussed the stabilization of a desired closed trajectory, named limit cycle, instead of a single point in order to achieve high steady state performance and meet the technological constraints [10]–[13]. However, the proposed controllers are developed by a deep analysis of the converter dynamics and the possibility of generalisation is not straightforward.

Theoretically, the authors in [14] have addressed the stabilization problem of a hybrid limit cycle consisting of two subsystems using a geometric approach. The limit cycle in question meets the control requirements and has been designed as close as possible to the desired state where necessary and sufficient conditions for its existence have also been discussed. In addition, a predictive approach has been proposed in [15] to guarantee a fast tracking of an optimized limit cycle. The latter represents the best cycle in the sense given by a criterion, which has been defined for reducing the quadratic norm of the tracking error. While Poincaré map approach is well-known as a powerful tool for stability analysis of limit cycles, it can also be used efficiently for switching law design as argued in [16] where a hybrid state feedback has been developed for a desired limit cycle stabilization of second order switched affine systems.

In this note, we aim to extend the results discussed in [16] and provide a systematic methodology for a desired limit cycle stabilization of switched affine systems. The main idea consists on finding the convenient switching surfaces that achieve the local asymptotic stability conditions reported in [17] for a predefined limit cycle. A multilevel DC/DC power converter has been considered to demonstrate the effectiveness of the investigated approach compared to the proposed one in [12].

Notations: I_n denotes the identity matrix with dimension n. $\mathcal{I}_N = \{1, 2, ..., N\}$ represents a finite index set with cardinal N. For $X \in \mathbb{R}^{n\times m}$, $\ker(X)$ denotes the null-space of X and $X^\perp$ represents its basis. $x \times y$ represents the cross product

M. Benmiloud is with LS2N, Ecole Centrale Nantes, 44300 Nantes, France. A. Benalia and M. Benmiloud (external member) are with LACoSERE, University of Amar Telidji, Laghouat (03000), Algeria ({med.benmiloud, a.benalia}@lagh-univ.dz). M. Djemai and M. Defoort are with LAMIH, UMR CNRS 8201, UVHC, 59313 Valenciennes, France. ({Mohamed.Djemai, Michael.Defoort}@univ-valenciennes.fr).

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2018.2841806, IEEE Transactions on Automatic Control

between vectors x and y.

II. Problem Statement

Consider the switched affine system:

$$\dot{x}(t) = A_{\sigma(t)}x(t) + B_{\sigma(t)} \tag{1}$$

where $x(t) \in \mathbb{R}^n$ is the state vector and $\sigma(t) : \mathbb{R}^+ \to \mathcal{I}_N$ is a piecewise constant and right-continuous function, which represents the switching law. This note addresses the switching law design for local stabilization of a desired closed trajectory, known as hybrid limit cycle, which is defined as follows.

Let us denote q_i the subsystem given by (1) for $\sigma(t) = i$, which will be called the discrete mode q_i. The continuous dynamics of the latter is given by (1) and can be written as follows:

$$\dot{x}(t) = f_{q_i}(x(t)) \tag{2}$$

Assume that system (1) admits a closed trajectory γ with period T. For simplicity and without loss of generality, we assume that this trajectory evolves consecutively from mode q_1, to mode q_2, and so on until it reaches mode q_k, and finally, after completing one cycle, it returns to mode q_1 as shown in Fig. 1. The periodic solution γ can be characterized by the following switching sequence [18]:

$$S_\gamma(x_0) = \{x_0, (t_1^*, q_1), (t_2^*, q_2), ..., (t_k^*, q_k)\} \tag{3}$$

where t_i^* is the spent time in mode q_i, x_0 is the initial state of the periodic trajectory with period $T = \sum_{i=1}^k t_i^*$.

The control objective corresponds to the design of a state-dependent switching law for the switched affine system (1) in order to guarantee the local asymptotic stability of the desired limit cycle defined by (3). To this end, the transition between successive discrete modes (q_i and q_{i+1}) in the discrete sequence (3) is governed by a switching surface, denoted $\mathcal{S}_{i,i+1}$, which corresponds to a hyperplane of dimension $n-1$ as follows:

$$\mathcal{S}_{i,i+1} = \{x \in \mathbb{R}^n : C_{i,i+1}^T \left(x - x_{i,i+1}^*\right) = 0\} \tag{4}$$

where $C_{i,i+1} \in \mathbb{R}^{n\times 1}$ is considered as a *design parameter*. The switching point $x_{i,i+1}^*$ corresponds to the intersection of the limit cycle trajectory with the switching surface $\mathcal{S}_{i,i+1}$ as shown in Fig. 1 with $x_0 = x_{k,1}^*$. It should be pointed out that the switching surface $\mathcal{S}_{i,i+1}$ is spanned by the orthogonal basis [19]:

$$C_{i,i+1}^{\perp} = \left[e_{i,i+1}^{(1)}\ e_{i,i+1}^{(2)}\ \cdots\ e_{i,i+1}^{(n-1)}\right] \tag{5}$$

where $(C_{i,i+1})^T C_{i,i+1}^{\perp} = \mathbf{0}_{n\times n-1}$.

By the parametrization of the switching law, given by (4), the control design problem can be reformulated as follows: *find the free parameters $C_{i,i+1}$ of each switching surface (orientation of different hyperplanes) in order to guarantee the local stability of the limit cycle γ.*

For this purpose, we recall in the next subsection the local stability analysis and stabilization of limit cycles for switched affine systems using hybrid Poincaré map approach. The latter will be used in Section III as an inverse problem to design the switching law.

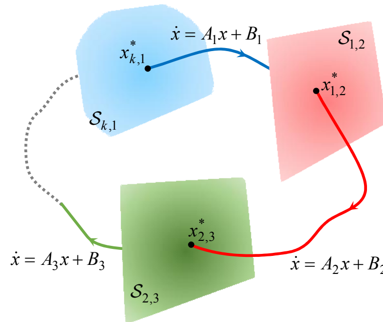

Fig. 1. Limit cycle with k discrete modes of a switched affine system.

A. Stability and Stabilization of Limit Cycles: Poincaré Map Approach

Existence and stability analysis of limit cycles and different type of oscillations can be determined using the Poincaré map approach [20]. Let us consider the limit cycle described above, the function defined for x in the neighborhood of $x_{k,1}^*$ as follows:

$$P(x) = P_{k,1} \circ ... \circ P_{2,3} \circ P_{1,2}(x) \tag{6}$$

is the Poincaré map extended to hybrid systems where function $P_{i,i+1}(x)$ associates to a given point $x_{i-1,i} \in \mathcal{S}_{i-1,i}$ a point $x_{i,i+1} \in \mathcal{S}_{i,i+1}$, which is the first intersection of the trajectory of mode q_i with the switching surface $\mathcal{S}_{i,i+1}$ [21]. Local stability of the limit cycle γ can be checked by looking at the eigenvalues position of the Jacobian of the map $P(x)$ in the complex plane with respect to a unit circle as follows [17]:

Theorem 1. *Consider the switched affine system (1)-(4). Assume there exists a limit cycle γ characterized by the switching sequence (3). Assume also that the limit cycle trajectory is transversal to the switching surfaces $\mathcal{S}_{1,2}, ..., \mathcal{S}_{k,1}$ at the switching points $x_{1,2}^*, ..., x_{k,1}^*$ respectively. The Jacobian of the map P is given by:*

$$dP = dP_{k,1}...dP_{3,4}dP_{2,3}dP_{1,2} \tag{7}$$

with

$$dP_{i,i+1} = \underbrace{\left(I_n - \frac{f_{q_i}(x_{i,i+1}^*)C_{i,i+1}^T}{C_{i,i+1}^T f_{q_i}(x_{i,i+1}^*)}\right)}_{\Psi_{i,i+1}} e^{A_i t_i^*} \tag{8}$$

The limit cycle γ is locally asymptotic stable if all the eigenvalues of dP are inside the unit circle. If at least one of the eigenvalues is outside the unit circle, then the limit cycle γ is unstable.

The transversality condition corresponds to $C_{i,i+1}^T f_{q_i}(x_{i,i+1}) \neq 0$, which guarantees the reachability of the switching surface by the trajectory of mode q_i at least locally.

The authors in [16] have discussed some interesting algebraic properties of the Jacobian of the map between mode q_i and mode q_{i+1}, denoted $dP_{i,i+1}$, for planar switched systems. The following proposition outlines the Jordan decomposition of the map $dP_{i,i+1}$.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2018.2841806, IEEE Transactions on Automatic Control

Proposition 1. *The Jacobian of the map between two successive modes, let us say mode q_i and mode q_{i+1}, given by (8) can be rewritten using the Jordan decomposition as follows:*

$$\Psi_{i,i+1} = V_{i,i+1}\Lambda V_{i,i+1}^{-1} \tag{9}$$

with $V_{i,i+1} = \left[f_{q_i}(x^*_{i,i+1})\, C^{\perp}_{i,i+1}\right] \in \mathbb{R}^{n\times n}$ and $\Lambda = diag(0,1,...,1) \in \mathbb{R}^{n\times n}$.

Based on this decomposition, conditions for switching surfaces design have been investigated in [16] for local stabilization of a desired limit cycle in 2^{nd} switched affine systems. The following theorem highlights this issue.

Theorem 2. *Assume the existence of a limit cycle with k discrete modes in a 2^{nd} order switched system given by (1)-(4). If one of the switching surfaces (switching lines) between two successive modes, let us say mode q_i and mode q_{i+1}, can be chosen tangent to the trajectory of mode q_{i+1} at the point $x^*_{i,i+1}$ as follows:*

$$C_{i,i+1} \in \ker\left(f^T_{q_{i+1}}(x^*_{i,i+1})\right) \tag{10}$$

then the limit cycle is locally asymptotic stable and all the eigenvalues of dP are equal to zero.

As the dimension of the switching surface $\mathcal{S}_{i,i+1}$ is equal to 1, condition (10) is equivalent to:

$$\mathcal{S}_{i,i+1} = \text{span}\{f^T_{q_{i+1}}(x^*_{i,i+1})\} \tag{11}$$

The authors in [16] have argued that the results of Theorem 2 can not be applied in the case where the trajectories of all successive discrete modes are parallel at the corresponding switching point of the limit cycle. In addition, if it is the case then the switching surfaces have no effect on the local stability of the limit cycle [16]. In the next section, we will extend the above results to a general case of n^{th} order switched affine systems. To this end, we assume that the following assumptions are fulfilled.

Assumption 1. *The vector fields of successive discrete modes are not collinear at the corresponding switching point of the limit cycle where the switching surfaces have to be designed.*

Assumption 2. *The number of switching surfaces (discrete modes), denoted k, is assumed to be greater than or equal to $n-1$.*

For notation convenience, we will use in the remainder of the paper V_i, Ψ_i and dP_i instead of $V_{i,i+1}$, $\Psi_{i,i+1}$ and $dP_{i,i+1}$, respectively.

III. Main results

Theorem 2 answers the question of how one can choose the design parameters in order to place the eigenvalues of dP at the origin. However, it should be noted that one of the eigenvalues of dP is always equal to zero due to slight variation analysis on the Poincaré section, which has a lower dimension.

Proposition 2. *Assume the existence of a limit cycle characterized by the discrete sequence (3). One of the eigenvalues of the Jacobian of the Poincaré map (dP) is equal to zero, which corresponds to the eigenvector $f_{q_1}(x^*_{k,1})$.*

Proof. Multiplying the right side of dP given by (7)-(8) with the vector $f_{q_1}(x^*_{k,1})$, one can find:

$$dPf_{q_1}(x^*_{k,1}) = \Psi_k e^{A_k t^*_k} ... \Psi_1 e^{A_1 t^*_1} f_{q_1}(x^*_{k,1})$$

From the fact that

$$f_{q_1}(x^*_{k,1}) = e^{-A_1 t^*_1} f_{q_1}(x^*_{1,2})$$

and taking into account that $f_{q_1}(x^*_{1,2})$ corresponds to the zero eigenvalue of Ψ_1 (Proposition 1), one can conclude that $f_{q_1}(x^*_{k,1})$ is an eigenvector for dP associated with a zero eigenvalue. □

The next theorem addresses the problem of how one can choose the switching surfaces to guarantee zero eigenvalues of dP and hence local asymptotic stability of the limit cycle for the general case of switched affine systems.

Theorem 3. *Assume the existence of a limit cycle with $k \geq n-1$ discrete modes for the switched system given by (1)-(4). If among the switching surfaces $n-1$ successive surfaces, let us say $\mathcal{S}_{1,2}$,..., $\mathcal{S}_{n-1,n}$ for simplicity, verify the following conditions:*

$$\mathcal{S}_{i,i+1} = span\{e^{(1)}_{i,i+1}, ..., e^{(n-i)}_{i,i+1}, e^{(n-i+1)}_{i,i+1}, ..., e^{(n-1)}_{i,i+1}\} \tag{12}$$

where:

$$\begin{cases} e^{(1)}_{i,i+1} &= f_{q_{i+1}}(x^*_{i,i+1}) \\ e^{(2)}_{i,i+1} &= e^{-A_{i+1}t^*_{i+1}} f_{q_{i+2}}(x^*_{i+1,i+2}) \\ e^{(3)}_{i,i+1} &= e^{-A_{i+1}t^*_{i+1}} e^{-A_{i+2}t^*_{i+2}} f_{q_{i+3}}(x^*_{i+2,i+3}) \\ \vdots & \\ e^{(n-i)}_{i,i+1} &= e^{-A_{i+1}t^*_{i+1}} \cdots e^{-A_{n-1}t^*_{n-1}} f_{q_n}(x^*_{n-1,n}) \end{cases} \tag{13}$$

and $e^{(n-i+1)}_{i,i+1}, ..., e^{(n-1)}_{i,i+1}$ are free vectors, such that $dim(\mathcal{S}_{i,i+1}) = n-1$, then the limit cycle is locally asymptotic stable and all the eigenvalues of dP are equal to zero.

Proof. The Jacobian of the Poincaré map corresponding to the discrete sequence of successive discrete modes : q_1, q_2, ... , q_n, can be written as below:

$$dP_n dP_{n-1} \cdots dP_1 = \Psi_n e^{A_n t^*_n} \Psi_{n-1} e^{A_{n-1}t^*_{n-1}} \cdots \Psi_1 e^{A_1 t^*_1}$$

From (13) and taking in mind the assumption that $dim(\mathcal{S}_{1,2}) = n-1$, the switching surface $\mathcal{S}_{1,2}$ is completely defined by the following $n-1$ linearly independent vectors:

$$\begin{cases} e^{(1)}_{1,2} &= f_{q_2}(x^*_{1,2}) \\ e^{(2)}_{1,2} &= e^{-A_2 t^*_2} f_{q_3}(x^*_{2,3}) \\ e^{(3)}_{1,2} &= e^{-A_2 t^*_2} e^{-A_3 t^*_3} f_{q_4}(x^*_{3,4}) \\ \vdots & \\ e^{(n-1)}_{1,2} &= e^{-A_2 t^*_2} \cdots e^{-A_{n-1}t^*_{n-1}} f_{q_n}(x^*_{n-1,n}) \end{cases} \tag{14}$$

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2018.2841806, IEEE Transactions on Automatic Control

However, the next switching surface $\mathcal{S}_{2,3}$ is not completely constrained. Indeed, from (13), it is spanned by the following $n-2$ linearly independent vectors:

$$\begin{cases} e_{2,3}^{(1)} & = f_{q_3}(x^*_{2,3}) \\ e_{2,3}^{(2)} & = e^{-A_3 t_3^*} f_{q_4}(x^*_{3,4}) \\ e_{2,3}^{(3)} & = e^{-A_3 t_3^*} e^{-A_4 t_4^*} f_{q_5}(x^*_{4,5}) \\ \vdots & \\ e_{2,3}^{(n-2)} & = e^{-A_3 t_3^*} \cdots e^{-A_{n-1} t_{n-1}^*} f_{q_n}(x^*_{n-1,n}) \end{cases} \tag{15}$$

together with a free linearly independent vector $e_{2,3}^{(n-1)}$. Recursively, only one basis vector for the last switching surface ($\mathcal{S}_{n-1,n}$) is restricted ($e_{n-1,n}^{(1)} = f_{q_n}(x^*_{n-1,n})$) and the remaining $n-2$ vectors are free and have to be linearly independent.

From Proposition 1 and the basis vectors (14) of the switching surface $\mathcal{S}_{1,2}$, one can find:

$$\begin{aligned} dP_2 dP_1 =& \Psi_2 e^{A_2 t_2^*} [\mathbf{0}_{n\times 1} \quad f_{q_2}(x^*_{1,2}) \quad e^{-A_2 t_2^*} f_{q_3}(x^*_{2,3}) \cdots \\ & e^{-A_2 t_2^*} e^{-A_3 t_3^*} \ldots e^{-A_{n-1} t_{n-1}^*} f_{q_n}(x^*_{n-1,n})] V_1^{-1} e^{A_1 t_1^*} \end{aligned}$$

By keeping in mind that $e^{A_2 t_2^*} f_{q_2}(x^*_{1,2}) = f_{q_2}(x^*_{2,3})$ is an eigenvector associated to a zero eigenvalue for the matrix Ψ_2, the above equation is simplified as follows:

$$\begin{aligned} dP_2 dP_1 =& [\mathbf{0}_{n\times 2} \quad \Psi_2 f_{q_3}(x^*_{2,3}) \quad \Psi_2 e^{-A_3 t_3^*} f_{q_4}(x^*_{3,4}) \cdots \\ & \Psi_2 e^{-A_3 t_3^*} \ldots e^{-A_{n-1} t_{n-1}^*} f_{q_n}(x^*_{n-1,n})] V_1^{-1} e^{A_1 t_1^*} \end{aligned}$$

From Proposition 1, the vectors given by (15) are the eigenvectors associated with the eigenvalue $\lambda = 1$ of the matrix Ψ_2. Hence, the matrix Ψ_2 can be omitted from the above equation.

$$\begin{aligned} dP_2 dP_1 =& [\mathbf{0}_{n\times 2} \quad f_{q_3}(x^*_{2,3}) \quad e^{-A_3 t_3^*} f_{q_4}(x^*_{3,4}) \cdots \\ & e^{-A_3 t_3^*} \ldots e^{-A_{n-1} t_{n-1}^*} f_{q_n}(x^*_{n-1,n})] V_1^{-1} e^{A_1 t_1^*} \end{aligned}$$

Multiplying this equality from the left by dP_3 and following the same steps as above yields:

$$\begin{aligned} dP_3 dP_2 dP_1 =& [\mathbf{0}_{n\times 3} \quad f_{q_4}(x^*_{3,4}) \quad e^{-A_4 t_4^*} f_{q_5}(x^*_{4,5}) \cdots \\ & e^{-A_4 t_4^*} \ldots e^{-A_{n-1} t_{n-1}^*} f_{q_n}(x^*_{n-1,n})] V_1^{-1} e^{A_1 t_1^*} \end{aligned}$$

With the same manner, the Jacobian of the hybrid Poincaré map corresponding to the sub-discrete sequence in question is given by:

$$dP_n dP_{n-1} \cdots dP_1 = [\mathbf{0}_{n\times n}] V_1^{-1} e^{A_1 t_1^*} = \mathbf{0}_{n\times n}$$

which yields $dP = \mathbf{0}_{n\times n}$. Hence, all the eigenvalues of dP are equal to zero and the limit cycle is locally asymptotic stable. □

It should be noted that Theorem 3 generalizes the proposed results for second order switched affine systems in Theorem 2. Indeed, condition (11) is equivalent to (12) where the switching surface from mode q_1 to mode q_2 is spanned by the vector field of mode q_2 at the switching point $x^*_{1,2}$ in order to achieve the local asymptotic stability of the limit cycle. However, one tangency is no longer sufficient to place all the eigenvalues of the Jacobian of the hybrid Poincaré map at the origin for switched affine systems with order bigger than 2. For example, the limit cycle of a third order switched affine system can be locally stabilized by an appropriate choice of two successive switching surfaces ($\mathcal{S}_{1,2}$ and $\mathcal{S}_{2,3}$ for simplicity) among the k switching surfaces as below:

$$\begin{cases} \mathcal{S}_{1,2} = & \text{span}\{f_{q_2}(x^*_{1,2}), e^{-A_2 t_2^*} f_{q_3}(x^*_{2,3})\} \\ \mathcal{S}_{2,3} = & \text{span}\{f_{q_3}(x^*_{2,3}), e_{2,3}^{(2)}\} \end{cases} \tag{16}$$

From the geometric point of view, the first basis vector $f_{q_2}(x^*_{1,2})$ ($f_{q_3}(x^*_{2,3})$) restricts the switching surface $\mathcal{S}_{1,2}$ ($\mathcal{S}_{2,3}$) to be a hyperplane tangent to the trajectory of the next discrete mode q_2 (q_3) at the switching point $x^*_{1,2}$ ($x^*_{2,3}$). Besides, the second basis vector of the switching surface $\mathcal{S}_{1,2}$ ($e^{-A_2 t_2^*} f_{q_3}(x^*_{2,3})$) restricts the vector field of mode q_3 at the switching point $x^*_{2,3}$ and moved backward the switching point $x^*_{2,3}$ by $e^{-A_2 t_2^*}$ to belong to the switching surface $\mathcal{S}_{1,2}$ as shown in Fig. 2 for a general notation case ($\mathcal{S}_{i,i+1}$ and $\mathcal{S}_{i+1,i+2}$).

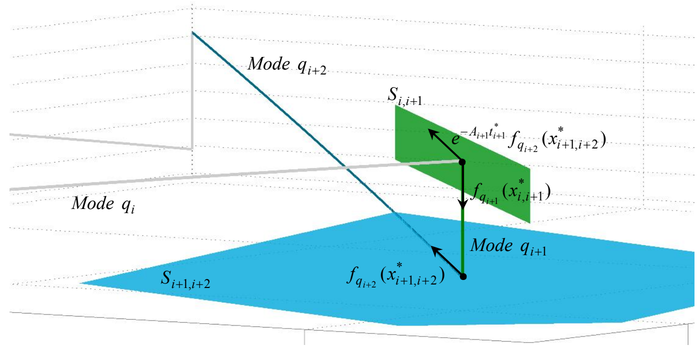

Fig. 2. Geometric illustration of stabilizing switching surfaces of a limit cycle of a third order switched affine system.

Remark 1. *For third order switched affine systems, the assumption that the vectors given by (13) are linearly independent is always verified. From (16), the first basis vector* $f_{q_2}(x^*_{1,2})$ *is equal to* $e^{-A_2 t_2^*} f_{q_2}(x^*_{2,3})$. *This means that the basis vector of* $\mathcal{S}_{1,2}$ *are linearly independent as* $f_{q_2}(x^*_{2,3})$ *and* $f_{q_3}(x^*_{2,3})$ *are not collinear (see Assumption 1).*

IV. Illustrative Example

The considered example is an application in power electronics known as multicellular converter. It belongs to the class of multilevel power converters, which represents an interesting option for powering various industrial applications with high dynamic performances [22]. Fig. 3 depicts the topology of three-cells chopper supplying an inductive load R-L. The converter structure is based on a series connection of commutation cells where each one is composed of pairs of complementary switches. The control signal associated with the k^{th} commutation cell is denoted u_k where $u_k = 1$ ($u_k = 0$) means that the upper switch is closed (opened) and the lower switch is opened (closed). Between each two successive commutation cells, a floating capacitor is inserted to split the input voltage (E) among different cells [22].

By considering the capacitors voltages (v_{c1} and v_{c2}) and the load current (i_L) as the continuous states, the converter dynamics can be described by a third order switched affine

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2018.2841806, IEEE Transactions on Automatic Control

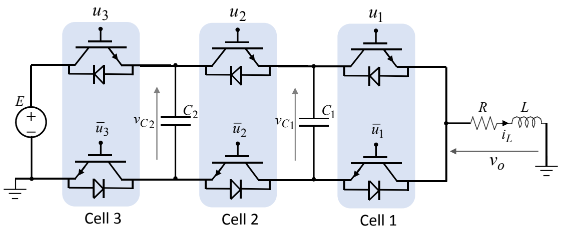

Fig. 3. Topology of a DC-DC three-cells converter supplying an inductive load.

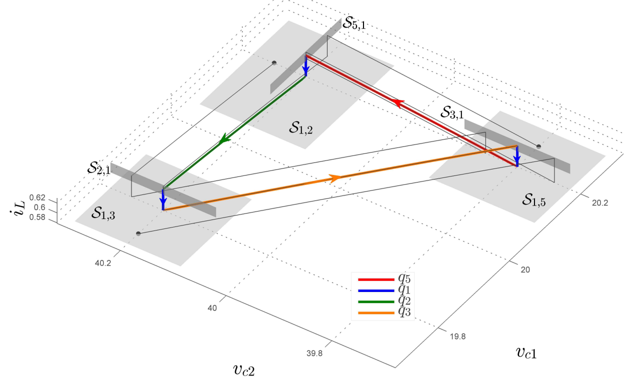

Fig. 4. The desired limit cycle of three-cells converter with the proposed switching surfaces in [12].

system with eight discrete modes. The latter corresponds to different input signals combinations (u_1, u_2, u_3) as follows: $q_1 \equiv (0,0,0)$, $q_2 \equiv (1,0,0)$, $q_3 \equiv (0,1,0)$, $q_4 \equiv (1,1,0)$, $q_5 \equiv (0,0,1)$, $q_6 \equiv (1,0,1)$, $q_7 \equiv (0,1,1)$, $q_8 \equiv (1,1,1)$. The state matrices of each discrete mode can be obtained by replacing the cell signals in the following matrices:

$$A_i = \begin{pmatrix} 0 & 0 & \frac{u_2-u_1}{C_1} \\ 0 & 0 & \frac{u_3-u_2}{C_2} \\ \frac{u_1-u_2}{L} & \frac{u_2-u_3}{L} & \frac{-R}{L} \end{pmatrix}, \; B_i = \begin{pmatrix} 0 \\ 0 \\ \frac{Eu_3}{L} \end{pmatrix}$$

The control objective is to force the converter trajectory to reach and maintain a desired state x_{ref}, which is not an equilibrium point of any discrete mode. Theoretically, this can be achieved only with infinitely fast switching among different discrete modes and this is not the case in practice due to the limited switching frequency of semiconductor power devices. To remedy this issue, the authors in [12] have discussed the hybrid control design for a desired limit cycle stabilization instead of a desired point. This limit cycle meets the technological constraints of the multicellular converter: switching frequency, dwell time and adjacency condition (only one switch can take place at each transition). However, the proposed switching surfaces have been designed by an analysis of the converter dynamics and followed by the stability analysis of the limit cycle. In this note, we will use the proposed control scheme in Theorem 3 for the local stabilization of a desired limit cycle, which will be presented in the next subsection.

A. Desired Limit Cycle Design

For converter safety, which corresponds to an equilibrated distribution of the input voltage constraint among different cells, the capacitor voltages have to be stabilized at the following average references $V_{ref_1} = E/3$ and $V_{ref_2} = 2E/3$. The load current have to be stabilized around the desired load current $I_{ref} \in [0, I_{max}]$ with $I_{max} = E/R$ is the maximum current that can be delivered by the converter.

Depending on the reference load current, three desired limit cycles have been distinguished in [12] where we will consider only one case to highlight the developed results. If $I_{ref} \in [0, \frac{1}{3}I_{max}]$ then the desired limit cycle has the following switching sequence:

$$S_{\gamma_1} = \{x^*_{5,1}, (t^*_1, q_1), (t^*_2, q_2), (t^*_1, q_1), (t^*_3, q_3), (t^*_1, q_1), (t^*_5, q_5)\} \quad (17)$$

where its geometry is illustrated in Fig. 4. The spent time in each discrete mode is given by: $t^*_1 = (\frac{1}{3} - \alpha)T$, $t^*_2 = t^*_3 = t^*_5 = \alpha T$ with $\alpha = \frac{I_{ref}}{I_{max}}$ and T represents the period of the limit cycle (see [12] for more details). The switching points of the desired limit cycle are:

$$x^*_{1,2} = \begin{pmatrix} V^+_{ref_1} \\ V^+_{ref_2} \\ I^-_{ref} \end{pmatrix}, \; x^*_{5,1} = \begin{pmatrix} V^+_{ref_1} \\ V^+_{ref_2} \\ I^+_{ref} \end{pmatrix}, \; x^*_{1,5} = \begin{pmatrix} V^+_{ref_1} \\ V^-_{ref_2} \\ I^-_{ref} \end{pmatrix}$$
$$x^*_{3,1} = \begin{pmatrix} V^+_{ref_1} \\ V^-_{ref_2} \\ I^+_{ref} \end{pmatrix}, \; x^*_{1,3} = \begin{pmatrix} V^-_{ref_1} \\ V^+_{ref_2} \\ I^-_{ref} \end{pmatrix}, \; x^*_{2,1} = \begin{pmatrix} V^-_{ref_1} \\ V^+_{ref_2} \\ I^+_{ref} \end{pmatrix} \quad (18)$$

where $V^\bullet_{ref_i} = V_{ref_i} \bullet \Delta v_i$ and $I^\bullet_{ref} = I_{ref} \bullet \Delta i$ with $\bullet$ corresponds to $+$ or $-$. The ripples of the capacitors voltages (Δv_1 and Δv_2) and the load current (Δi) are defined below:

$$\begin{cases} \Delta v_1 = \left(\frac{I_{ref}}{C_1}\right)\frac{\alpha T}{2}, \; \Delta v_2 = \left(\frac{I_{ref}}{C_2}\right)\frac{\alpha T}{2} \\ \Delta i = \alpha\left(\frac{1}{3} - \alpha\right)\frac{ET}{2L} \end{cases} \quad (19)$$

In this case, the desired limit cycle of three-cells converter is completely defined for the stabilization purpose where one of the parameters T, Δv_1, Δv_2 or Δi can be fixed and the others can be calculated using (19).

B. Control Design

By a deep analysis of the converter dynamics, the authors in [12] have proposed the following design parameters for each switching surface:

$$C_{1,2} = [0\,0\,1]^T, \; C_{2,1} = [1\,0\,0]^T, \; C_{1,3} = [0\,0\,1]^T$$
$$C_{3,1} = [1\,0\,0]^T, \; C_{1,5} = [0\,0\,1]^T, \; C_{5,1} = [0\,1\,0]^T$$

From Theorem 3, we are looking for a sub-discrete sequence with three discrete modes where the two switching surfaces between them verify conditions given by (16). One can distinguish six sub-discrete sequences to be analyzed: (q_1, q_2, q_1), (q_2, q_1, q_3), (q_1, q_3, q_1), (q_3, q_1, q_5), (q_1, q_5, q_1) and (q_5, q_1, q_2). Let us consider the first switching surface in the sub-discrete sequence (q_1, q_2, q_1). From (16), we have:

$$\mathcal{S}_{1,2} = \text{span}\{f_{q_2}(x^*_{1,2}), e^{-A_2 t^*_2} f_{q_1}(x^*_{2,1})\}$$

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2018.2841806, IEEE Transactions on Automatic Control

The vectors $f_{q_2}(x^*_{1,2})$ and $e^{-A_2 t^*_2} f_{q_1}(x^*_{2,1})$ are linearity independent and the normal vector of the switching surface ($C_{1,2}$) can be computed by a cross product between them. Using the converter parameters reported in Table I, one can get the normalized design parameter $C_{1,2} = [0 \ \ -0.902 \ \ 0]^T$. The obtained switching surface does not meet the transversality condition ($C^T_{1,2} f_{q_1}(x^*_{1,2}) = 0$). Indeed, the vector field of mode q_1 at the switching point $x^*_{1,2}$ belongs to the switching surface $\mathcal{S}_{1,2}$ and it is not transversal. Hence, the sub-discrete sequence in question can not be used to stabilize the limit cycle. This is also true for the sub-discrete sequences: (q_1, q_3, q_1) and (q_1, q_5, q_1).

However, the remaining sub-discrete sequences (q_2, q_1, q_3), (q_3, q_1, q_5) and (q_5, q_1, q_2) verify the conditions of Theorem 3. Following the same steps as above, one can get the following normalized design parameters for each switching surface:

Sub-discrete sequence (q_2, q_1, q_3): From (16), we have:

$$\begin{cases} \mathcal{S}_{2,1} = & \text{span}\{f_{q_1}(x^*_{2,1}), e^{-A_1 t^*_1} f_{q_3}(x^*_{1,3})\} \\ \mathcal{S}_{1,3} = & \text{span}\{f_{q_3}(x^*_{1,3}), e^{(2)}_{1,3}\} \end{cases}$$

The choice of the basis vector $e^{(2)}_{1,3}$ is free and does not affect the eigenvalues of the matrix dP. In order to guarantee a minimal projection of $f_{q_1}(x^*_{1,3})$ on the hyperplane $\mathcal{S}_{1,3}$ and ensures the condition $C^T_{1,3} f_{q_1}(x^*_{1,3}) \neq 0$, the other basis vector can be chosen as follows:

$$e^{(2)}_{1,3} = f_{q_1}(x^*_{1,3}) \times f_{q_3}(x^*_{1,3})$$

Numerically, one can get:

$$C_{2,1} = [-0.707 - 0.707 \ 0]^T, \ C_{1,3} = [0.061 - 0.061 - 0.996]^T$$

With the same manner, one can obtain the following results:

$$C_{3,1} = [1 \ 0 \ 0]^T, \ C_{1,5} = [0 \ 0.118 \ \ -0.993]^T$$
$$C_{1,2} = [-0.118 \ 0 \ \ -0.993]^T, \ C_{5,1} = [0 \ 1 \ 0]^T$$

Fig. 5 illustrates the investigated switching surfaces for local stabilization of the desired limit cycle γ_1. One can remark that the converter trajectory converges to the desired limit cycle from different initial conditions ($x_0 \in \mathcal{S}_{1,2}$, $x_0 \in \mathcal{S}_{1,3}$ and $x_0 \in \mathcal{S}_{1,5}$) after hitting two switching surfaces. This is justified by the fact that each pair of the switching surfaces: ($\mathcal{S}_{2,1}$, $\mathcal{S}_{1,3}$), ($\mathcal{S}_{3,1}$, $\mathcal{S}_{1,5}$) and ($\mathcal{S}_{5,1}$, $\mathcal{S}_{1,2}$) assesses zero eigenvalues for the Jacobian of the Poincaré map (dP). However, the proposed switching surfaces in [12], shown in Fig. 4, guarantee the convergence of the converter trajectory to the desired limit cycle at least after hitting three surfaces (five switching surfaces in the case where $x_0 \in \mathcal{S}_{1,2}$). In summary, the investigated results offer a systematic approach for control law design dedicated to the local stabilization of a predefined limit cycle in switched affine systems.

TABLE I
A TYPICAL DC-DC THREE-CELLS CONVERTER PARAMETERS.

Parameter	E	C_1, C_2	L	R	I_{ref}
Value	$60V$	$40\mu F$	$5mH$	20Ω	$0.6A$

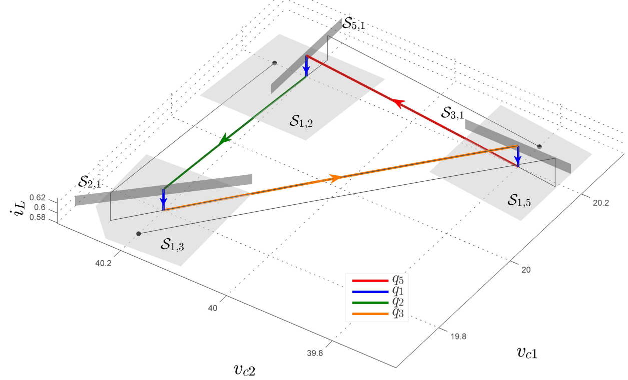

Fig. 5. The desired limit cycle of three-cells converter with the investigated switching surfaces.

V. CONCLUSION

In this note, we have tackled the problem of switching law design for local stabilization of a desired limit cycle for switched affine systems. The derived switching surfaces are obtained by an algebraic analysis of their effect on the eigenvalues of the Jacobian of the Poincaré map and hence on the local stability of the limit cycle. An illustrative example is provided to show the effectiveness and the simplicity of the proposed approach. As a future work, the robustness of the proposed control strategy has to be addressed in a rigourous manner. Another issue has to be also considered which corresponds to the characterization of the local stability region of the limit cycle and how it can be maximized. Besides, the proposed approach is limited to the case where the number of discrete modes k is bigger or equal to $n-1$ in order to be able to place all the eigenvalues of dP at the origin. To overcome this situation, adaptive switching surfaces (per cycle) can be addressed.

REFERENCES

[1] Z. Sun and S. S. Ge, *Stability theory of switched dynamical systems*. Springer Science and Business Media, 2011.
[2] D. Liberzon, *Switching in Systems and Control*. Birkhuser, 2003.
[3] L. Hetel and E. Bernuau, "Local stabilization of switched affine systems," *IEEE Transactions on Automatic Control*, pp. 1158 – 1163, Mar 2015.
[4] D. Kamri, R. Bourdais, J. Buisson, and C. Larbes, "Practical stabilization for piecewise-affine systems: A BMI approach," *Nonlinear Analysis: Hybrid Systems*, vol. 6, no. 3, pp. 859 – 870, 2012.
[5] H. Lin and P. J. Antsaklis, "Stability and stabilizability of switched linear systems: A survey of recent results," *IEEE Transactions on Automatic Control*, vol. 54, no. 2, pp. 308–322, Feb 2009.
[6] J. L. Mancilla-Aguilar and R. A. Garca, "Some results on the stabilization of switched systems," *Automatica*, vol. 49, no. 2, pp. 441 – 447, 2013.
[7] P. Bolzern and W. Spinelli, "Quadratic stabilization of a switched affine system about a nonequilibrium point," in *Proc American Control Conference*, vol. 5, June 2004, pp. 3890–3895.
[8] C. Seatzu, D. Corona, A. Giua, and A. Bemporad, "Optimal control of continuous-time switched affine systems," *IEEE Transactions on Automatic Control*, vol. 51, no. 5, pp. 726–741, May 2006.
[9] L. Zhang, S. Zhuang, and R. D. Braatz, "Switched model predictive control of switched linear systems: Feasibility, stability and robustness," *Automatica*, vol. 67, pp. 8 – 21, 2016.
[10] C. Sreekumar and V. Agarwal, "A hybrid control algorithm for voltage regulation in dc/dc boost converter," *IEEE Transactions on Industrial Electronics*, vol. 55, no. 6, pp. 2530–2538, June 2008.

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2018.2841806, IEEE Transactions on Automatic Control

[11] M. Benmiloud and A. Benalia, "Finite-time stabilization of the limit cycle of two-cell dc/dc converter: hybrid approach," *Nonlinear Dynamics*, vol. 83, pp. 319–332, 2015.

[12] M. Benmiloud, A. Benalia, M. Defoort, and M. Djemai, "On the limit cycle stabilization of a dc/dc three-cell converter," *Control Engineering Practice*, vol. 49, pp. 29 – 41, 2016.

[13] M. Bougrine, M. Benmiloud, A. Benalia, E. Delaleau, and M. Benbouzid, "Load estimator-based hybrid controller design for two-interleaved boost converter dedicated to renewable energy and automotive applications," ISA *Transactions*, vol. 66, pp. 425–436, 2017.

[14] J. Ben Salah, C. Valentin, H. Jerbi, and C. Xu, "Geometric synthesis of a hybrid limit cycle for the stabilizing control of a class of nonlinear switched dynamical systems," *Systems and Control Letters*, vol. 60, no. 12, pp. 967 – 976, 2011.

[15] D. Patino, P. Riedinger, and F. Ruiz, "A predictive control approach for dc-dc power converters and cyclic switched systems," in *Proceeding of the IEEE International Conference on Industrial Technology (ICIT)*, March 2010, pp. 1259–1264.

[16] M. Benmiloud, A. Benalia, M. Djemai, and M. Defoort, "Hybrid control design for limit cycle stabilization of planar switched systems," *International Journal of Control*, pp. 1–16, 2017.

[17] J. Gonçalves, "Regions of stability for limit cycle oscillations in piecewise linear systems," *IEEE Transactions on Automatic Control*, vol. 50, no. 11, pp. 1877–1882, Nov 2005.

[18] Z. Sun and S. S. Ge, *Switched Linear Systems: Control and Design*. Springer-Verlag London, 2005.

[19] M. Rubensson and B. Lennartson, "Stability of limit cycles in hybrid systems using discrete-time lyapunov techniques," in *Proceeding of 39th IEEE Conference on Decision and Control*, 2000, pp. 1397–1402.

[20] T. Parker and L. Chua, *Practical Numerical Algorithms for Chaotic Systems*. Springer New York, 1989.

[21] A. Girard, "Computation and stability analysis of limit cycles in piecewise linear hybrid systems," in *1st IFAC Conference on Analysis and Design of Hybrid Systems*, 2003, pp. 181–186.

[22] G. Gateau, M. Fadel, P. Maussion, R. Bensaid, and T. A. Meynard, "Multicell converters: active control and observation of flying-capacitor voltages," *IEEE Transactions on Industrial Electronics*, vol. 49, no. 5, pp. 998–1008, Oct 2002.